How Does AI Work?

Part 2: Neural Networks

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Review of Previous Presentation

Previous Presentation on How Does AI Work? Part 1: Pattern Recognition here.

Input → Machine Learning Model → Output.

Machine Learning models utilize pattern recognition to predict and classify data inputs.

A simple type of pattern recognition is Nearest Neighbor, where input is categorized based on its proximity to existing data points.

Neural Networks

Neural Networks are fundamentally **still a pattern recognition** system.

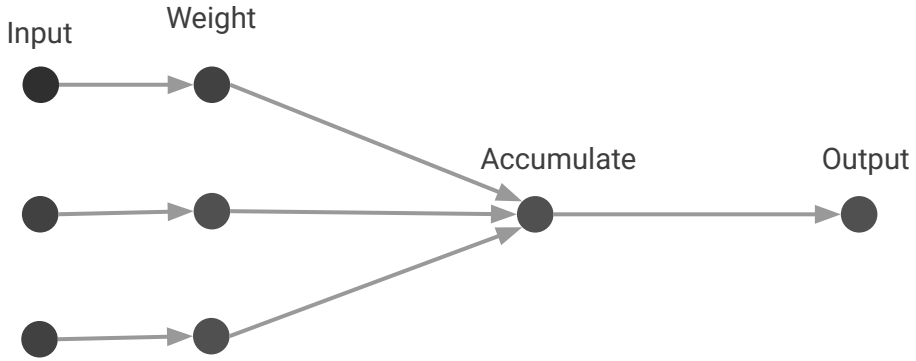
Neural Networks (cont.)

- A neural network is a type of ML model that consists of a set of connected nodes where each node assigns weights to its inputs.

Node: a basic building block of neural networks that assigns weights to inputs.

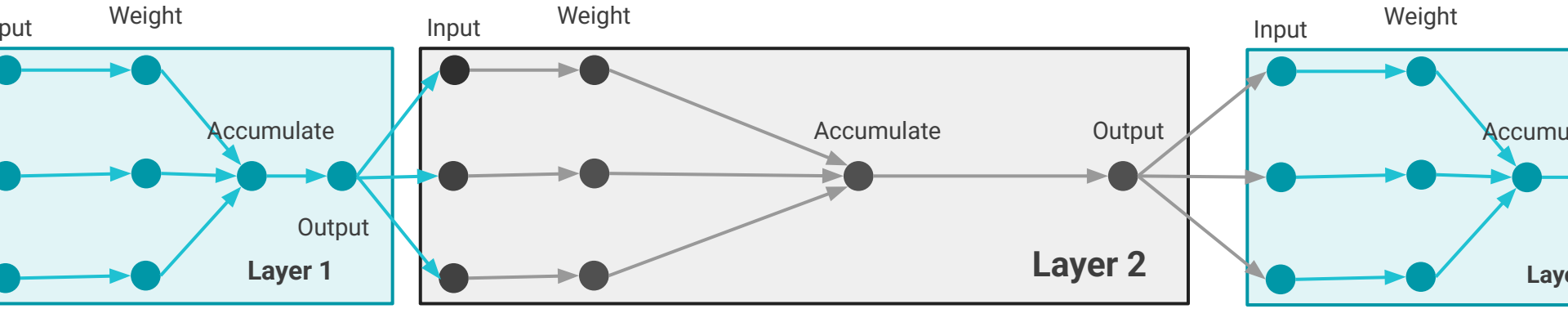
Weight: a set of numbers that make up the model.

E.g. list of words/sentences in the contract



From Neural Networks to Deep Learning

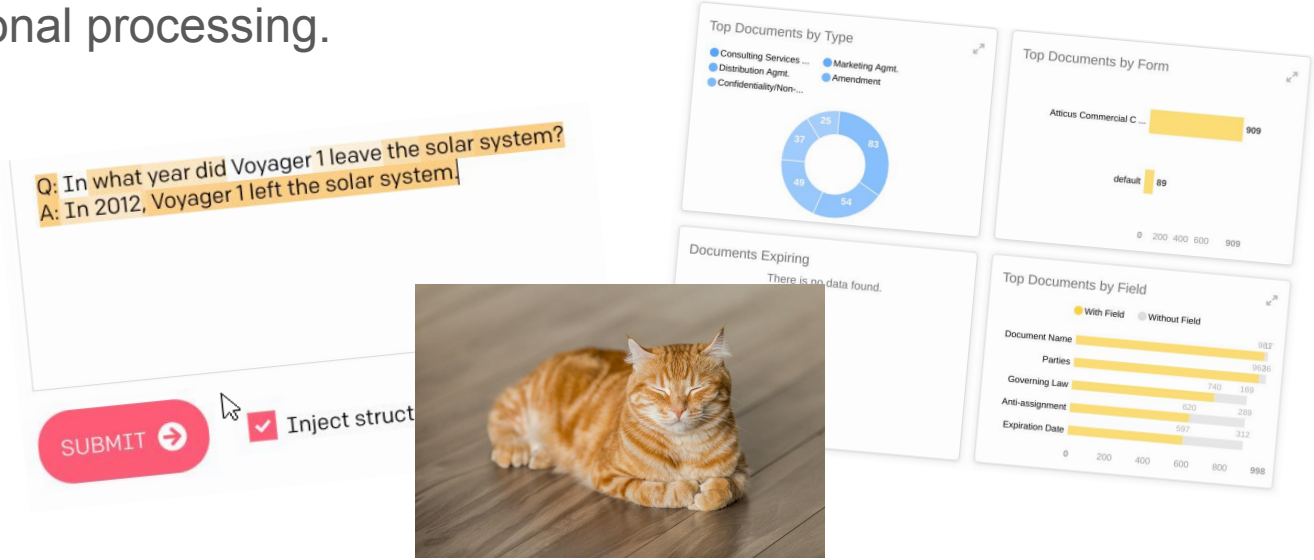
- "Deep" learning refers to a type of ML model that uses layers of neural networks.
- This allows it to make more complex decisions, like speech-to-text, sentiment detection, and text summarization.
- Nowadays, almost all models have multiple layers, and some have hundreds of layers.



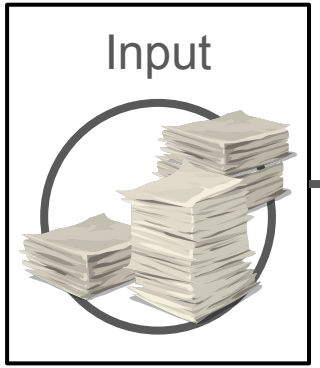
Output

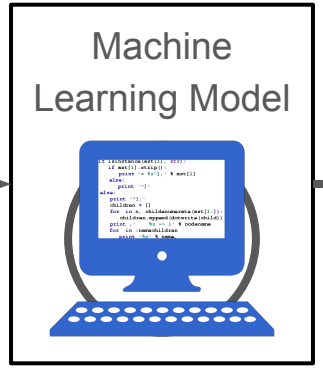
- The AI will then produce an output in a format that humans can understand.
  - A simple output would be the probability of a clause being a Change of Control clause.

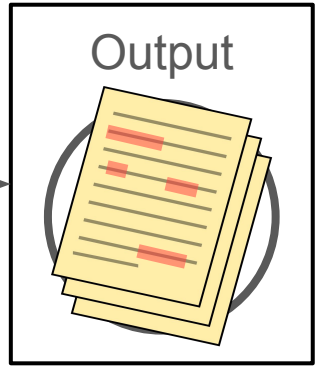
- More advanced outputs, such as long text, images, and graphs, etc., require additional processing.



Review of Steps







- Pre-processing, usually done using a simpler program.

- Based on Pattern Recognition
- Neural Networks: made of systems of nodes, communicating with each other
- Deep Learning: Neural Networks with multiple layers of processing

- AI produces an output humans can use and understand.